GREEN SHIELD

Sustainable Active Packaging Film

What Makes Green Shield Different?

Green Shield combines agricultural waste, biodegradable film development, and plant-derived antimicrobial properties in one sustainable packaging solution.

Key Features

1. Environmental Sustainability: Offers a potential alternative to conventional petroleum-based plastic packaging.
2. Agricultural Waste Utilization: Converts corn cobs, orange peels, and other agricultural residues into useful packaging materials.
3. Biodegradable Film Development: Uses starch and cellulose to develop films as alternatives to conventional plastic packaging.
4. Natural Active Packaging: Incorporates plant extracts to investigate antimicrobial properties against selected microorganisms.
5. Food Preservation: Aims to reduce microbial contamination and food spoilage, potentially extending shelf life.
6. Material Optimization: Evaluates strength, flexibility, thickness, moisture resistance, biodegradability, and antimicrobial activity.
7. Sustainability and Cost: Explores waste-to-value production and compares environmental impacts with conventional packaging.
8. Future Scope: Supports further research, pilot-scale production, and potential commercial applications.

Our Vision: To develop sustainable packaging by combining agricultural waste valorization, biodegradable materials, and plant-derived functionality.

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Step-by-Step Film Methodology

From natural resources → to a sustainable future

Raw Materials



Cornstarch



Glycerol



Pectin



Acetic acid



Neem extract



Distilled water

1 PREPARE PECTIN GEL



2 PREPARE STARCH MIXTURE



3 COMBINE & DISSOLVE



4 HEAT



5 JUST BEFORE REMOVING FROM HEAT



6 REMOVE FROM HEAT



7 ADD ACETIC ACID



8 CAST THE FILM



9 DRY



10 FINAL FILM



Your Standard Procedure